

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Fourth Semester of B. Pharm. Examination
University Theory Examination May 2016
PH225 Physical Pharmaceutics

Date: 06/05/2016, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Methods for determining particle size include;
[A] Optical microscopy
[B] Sieving
[C] Sedimentation
[D] All of these
 2. The particle size in sub sieve range can be found by gravity sedimentation as expressed in.....
[A] Van't hoff factor
[B] Ohm's law
[C] Graham's law
[D] Stoke's law
 3. Molecular dispersion ranges from.....
[A] 1 to 500nm
[B] < 1nm
[C] < 500nm
[D] None of these
 4. Scattering of light from colloidal particles is
[A] van't hoff factor
[B] faraday Tyndall effect
[C] van't hoff dispersion
[D] none
 5. Millimicron is commonly known as
[A] nanometer
[B] micrometer
[C] millimeter
[D] meter
 6. Viscosity of gaseous material _____ on increasing of temperature.
[A] increase
[B] decrease
[C] unaffected
[D] none

7. Substances possessing dilatant flow properties are suspensions containing _____ of small, deflocculated particles.
 [A] 50% or more
 [B] 10% to 20%
 [C] 20% to 40%
 [D] 30% to 50%
8. Bingham bodies begins to flow when shearing stress exceeds
 [A] yield value
 [B] pie value
 [C] vero value
 [D] eta value
9. Viscosity is an expression of the
 [A] resistance to flow
 [B] assistance to flow
 [C] resonance to flow
 [D] desiment to flow
10. To improve flow characteristics, material termed _____ are frequently added to granular powder.
 [A] glidants
 [B] retardants
 [C] tanants
 [D] mutants
11. Which of the following is a type of attractive intermolecular force?
 [A] Van der Waals Force
 [B] Ion-Dipole and Ion-Induced Dipole Force
 [C] Hydrogen Bond
 [D] All
12. Solids having definite shape and **orderly arrangement of units is called as** _____.
 [A] Amorphus
 [B] Critol
 [C] Ampicrysp
 [D] Crystal
13. Common types of thermal analysis is _____.
 [A] differential scanning calorimetry
 [B] differential thermal analysis
 [C] thermogravimetric analysis
 [D] all of above
14. When a solubility of non polar liquid in water is increaesed by surface active agent, **the phenomena is called as** _____.
 [A] ionic solubilization
 [B] partial solubilization
 [C] neo solubilization
 [D] micellar solubilisation
15. Solubility of weak electrolytes is strongly influenced by _____ of the solution.
 [A] temperature
 [B] pressure
 [C] osmolarity
 [D] pH

16. Erratic movement of particles of dispersion medium is called as
 [A] erratic motion
 [B] Tyndall effect
 [C] effective motion
 [D] Brownian motion
17. Parts of solvent required for 1 part of very soluble solid is _____
 [A] less than 1 part
 [B] less than 10 part
 [C] less than 5 part
 [D] less than 20 part
18. Solution in which solute phase is in equilibrium with solvent phase is called as _____
 [A] saturated solution
 [B] supersaturated solution
 [C] unsaturated solution
 [D] non saturated solution
19. Volume of the spaces in the powder sample is known as
 [A] void volume
 [B] void volume
 [C] void volume
 [D] void volume
20. Density is universally defined as
 [A] volume per unit weight
 [B] weight per unit volume
 [C] weight per unit weight
 [D] volume per unit volume

SECTION – II

Q 2 Attempt any **FOUR** of the following

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|----------|---|-----------|
| A | Define following terms: | 05 |
| | 1. Latent heat of vaporization | |
| | 2. Eutectic mixtures | |
| | 3. Buffer capacity | |
| | 4. Isotonic solution | |
| | 5. Colloidal dispersion | |
| B | Differentiate between flocculated and deflocculated suspension. | 05 |
| C | Write note on liquid crystals. | 05 |
| D | Enlist the types of colloids. Explain any one in detail. | 05 |
| E | Write note on optical properties of colloids. | 05 |
| F | Enlist methods of adjustment of isotonicity. Explain any one. | 05 |

SECTION – III

- Q 3** Attempt any **FOUR** of the following
- A** Define micromeritics. Enumerate different commonly used methods for measurement of particle size determination. Describe any one in detail. 05
 - B** Name the two fundamental properties of powders. Explain the application of derived properties of powders in pharmacy. 05
 - C** Explain the particle volume measurement by coulter counter method. 05
 - D** What is spreading coefficient? Derive its equation. 05
 - E** Write a note on surface active agents. 05
 - F** Discuss the electric properties of interfaces. 05
- Q 4** Attempt any **FOUR** of the following
- A** Give the working principle of cup and bob viscometer with a labelled diagram. 05
 - B** Explain non Newtonian type of flow with rheograms, mechanism and suitable examples. 05
 - C** Write a short note on Gel-Sol-Gel transformation. 05
 - D** Explain any binding forces between molecules. 05
 - E** Classify types of complex. Explain inclusion complex in detail. 05
 - F** Define and classify emulsions. Explain inbrief physical stability of emulsion. 05
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